

Power BI Semantic Model

Complete Technical & Functional DAX Measures Documentation

TOTAL MEASURES

46 Measures

DEDICATED TABLE

_Measures

DISPLAY FOLDERS

5 Structured Areas

DOMAIN AREA

Rail Operations & Sales

Executive Summary: This technical reference catalog provides full functional and DAX architectural documentation for all **46 measures** contained in the _Measures table of the Power BI semantic model. The model is structured into 5 logical display folders supporting executive reporting, passenger behavior analytics, revenue & sales performance, operational punctuality, and route corridor dynamics.

Semantic Model Structure & Folders

00 Common (9 Measures)

Total Revenue	£#,###0
Ticket Count	#,###0
Total Revenue (by Purchase Date)	£#,###0
Average Ticket Price	£#,###0.00
Revenue Previous Month	£#,###0
Revenue MoM %	0.0%; -0.0%
Ticket Count MoM %	0.0%; -0.0%
Revenue % of Total	0.0%
Journey Days	#,###0

01 Passenger Analysis (9 Measures)

Railcard Tickets	#,###0
Railcard Penetration %	0.0%
Avg Price (Railcard Holders)	£#,###0.00
Avg Price (No Railcard)	£#,###0.00
First Class Mix %	0.0%
Advance Booking %	0.0%
Online Purchase %	0.0%
Contactless Payment %	0.0%
Peak Travel Band	String

02 Sales Performance (8 Measures)

Avg Daily Revenue	£#,###0
Best Day Revenue	£#,###0
Online Revenue	£#,###0
Station Revenue	£#,###0
Online Revenue Mix %	0.0%
Revenue MTD	£#,###0
Avg Purchase Lead Time (Days)	0.0
Same Day Purchase %	0.0%

03 Railway Performance (12 Measures)

On Time Journeys / On Time %	#,###0 / 0.0%
Delayed Journeys / Delayed %	#,###0 / 0.0%
Cancelled Journeys / Canc. %	#,###0 / 0.0%
Avg Delay Minutes / Total Delay	0.0 / #,###0
Refund Requests / Refund Rate %	#,###0 / 0.0%
Revenue at Risk (Refunds)	£#,###0
Top Delay Reason	String

04 Route Analysis (8 Measures)

Distinct Routes	#,###0	Revenue by Arrival Station	£#,###0
Avg Tickets per Route	#,###0.0	Tickets by Arrival Station	#,###0
Avg Revenue per Route	£#,###0	Top Departure Station	String
Route Yield per Ticket	£#,###0.00	Busiest Route	String

SCHEMA

Semantic Model Architecture & Relationships

The semantic model follows a Star Schema centered on the FACT_TICKET_SALES fact table with multiple conforming dimension tables and role-playing relationships:

Dimension Table	Primary / Foreign Key	Active Status	Analytical Role & Role-Playing Context
DIM_DATE	FACT_TICKET_SALES[JourneyDateKey]	Active	Default temporal filter based on <i>Journey Date</i> (when travel occurred).
DIM_DATE (Role-Playing)	FACT_TICKET_SALES[PurchaseDateKey]	Inactive	Activated via USERRELATIONSHIP for booking date analysis (e.g. [Total Revenue (by Purchase Date)]).
DIM_STATION	FACT_TICKET_SALES[DepartureStationKey]	Active	Default geographic filter based on the <i>Origin / Departure Station</i> .
DIM_STATION (Role-Playing)	FACT_TICKET_SALES[ArrivalStationKey]	Inactive	Activated via USERRELATIONSHIP for destination station metrics ([Revenue by Arrival Station]).
DIM_TIME	FACT_TICKET_SALES[DepartureTimeKey]	Active	Time-of-day analytics and departure time banding (e.g. Morning Peak, Evening Peak).
DIM_PURCHASE_CHANNEL	FACT_TICKET_SALES[ChannelKey]	Active	Channel analytics (Online vs Station, Payment Methods, Railcard types).
DIM_TICKET_TYPE	FACT_TICKET_SALES[TicketTypeKey]	Active	Ticket class (Standard vs First Class) and fare type (Advance, Anytime, Off-Peak).
DIM_JOURNEY_STATUS	FACT_TICKET_SALES[StatusKey]	Active	Punctuality metrics (On Time, Delayed, Cancelled) and reason for disruption.

00 Common

Foundational base measures, time-intelligence calculations, volume counters, and company-wide financial KPIs.

Total Revenue

00 Common

£#,###0

Business Purpose: Calculates total gross revenue generated from ticket sales across all completed transactions. Serves as the primary financial base measure across the entire reporting solution.

```
SUM('FACT_TICKET_SALES'[Price])
```

Data Type: Currency / Integer

Referenced By: Avg Ticket Price, Revenue MoM %, Revenue % of Total, Avg Daily Revenue

Ticket Count

00 Common

#,###0

Business Purpose: Computes the total number of ticket sales transactions (passenger journeys booked). Forms the primary denominator for rate calculations, mixes, and unit averages.

```
COUNTROWS('FACT_TICKET_SALES')
```

Data Type: Integer

Referenced By: Average Ticket Price, Ticket Count MoM %, On Time %, Delayed %, Advance Booking %

Total Revenue (by Purchase Date)

00 Common

£#,###0

Business Purpose: Evaluates total sales revenue aligned to the *Purchase Date* (cash collection/booking date) rather than the default travel date. Essential for cash flow analysis and financial booking reconciliation.

```
CALCULATE(
    [Total Revenue],
    USERELATIONSHIP('FACT_TICKET_SALES'[PurchaseDateKey], DIM_DATE[DateKey])
)
```

Role-Playing Mechanism: USERELATIONSHIP on PurchaseDateKey

Active Dimension Swapped: Switches from JourneyDateKey to PurchaseDateKey

Average Ticket Price

00 Common

£#,##0.00

Business Purpose: Calculates the average realized fare per ticket sold (Average Order Value / Yield per transaction). Uses safe division to prevent divide-by-zero errors when filtered.

```
DIVIDE([Total Revenue], [Ticket Count])
```

Dependencies: [Total Revenue], [Ticket Count]

Referenced By: Avg Price (Railcard Holders), Avg Price (No Railcard)

Revenue Previous Month

00 Common

£#,##0

Business Purpose: Time-intelligence measure retrieving total ticket revenue for the preceding calendar month based on travel date. Powers month-over-month baseline comparisons.

```
CALCULATE([Total Revenue], PREVIOUSMONTH(DIM_DATE[Date]))
```

Time Intelligence Function: PREVIOUSMONTH over DIM_DATE[Date]

Referenced By: Revenue MoM %

Revenue MoM %

00 Common

0.0%;-0.0%;0.0%

Business Purpose: Calculates the percentage growth or contraction in revenue compared to the prior calendar month. Formatted with positive, negative, and zero display rules.

```
DIVIDE([Total Revenue] - [Revenue Previous Month], [Revenue Previous Month])
```

Formula Logic: (Current Rev - Prior Rev) / Prior Rev

Format String: Explicit positive / negative format: 0.0%;-0.0%;0.0%

Ticket Count MoM %

00 Common

0.0%;-0.0%;0.0%

Business Purpose: Measures the percentage change in passenger volume (tickets sold) month-over-month. Employs a DAX variable (VAR Prev) for clean, performant evaluation.

```
VAR Prev = CALCULATE([Ticket Count], PREVIOUSMONTH(DIM_DATE[Date]))
RETURN DIVIDE([Ticket Count] - Prev, Prev)
```

Optimization: Stores previous month ticket count in VAR to avoid re-evaluating

Format: Percentage with 1 decimal place

Revenue % of Total

00 Common

0.0%

Business Purpose: Calculates the proportion of total network revenue represented by the currently active filter context (e.g., share of revenue generated by a specific station, ticket type, or channel).

```
DIVIDE(
    [Total Revenue],
    CALCULATE([Total Revenue], REMOVEFILTERS())
)
```

Filter Modifier: REMOVEFILTERS() clears all slicer and visual filters for denominator

Use Case: Pareto charts, market share breakdown, category mix

Journey Days

00 Common

#,##0

Business Purpose: Counts the distinct number of calendar days on which journeys occurred within the filtered context. Serves as the denominator for daily rate calculations.

```
DISTINCTCOUNT('FACT_TICKET_SALES'[JourneyDateKey])
```

Column Counted: FACT_TICKET_SALES[JourneyDateKey]

Referenced By: Avg Daily Revenue

01 Passenger Analysis

Passenger demographics, concession railcard usage, class selection, purchasing channels, payment mechanisms, and peak travel demand.

Railcard Tickets

01 Passenger Analysis

#,##0

Business Purpose: Filters and counts all tickets sold where a promotional or concession railcard discount was applied (excluding non-discounted tickets marked as "None").

```
CALCULATE([Ticket Count], DIM_PURCHASE_CHANNEL[Railcard] <> "None")
```

Filter Condition: DIM_PURCHASE_CHANNEL[Railcard] <> "None"

Referenced By: Railcard Penetration %

Railcard Penetration %

01 Passenger Analysis

0.0%

Business Purpose: Measures the market share percentage of tickets purchased using railcards versus total tickets. Evaluates discount loyalty adoption.

```
DIVIDE([Railcard Tickets], [Ticket Count])
```

Calculation: Railcard Tickets / Total Tickets

Target Audience: Commercial & Marketing Teams

Avg Price (Railcard Holders)

01 Passenger Analysis

£#,##0.00

Business Purpose: Calculates the average ticket price paid specifically by passengers who applied a railcard discount. Used to benchmark discounted ticket yields against full-fare tickets.

```
CALCULATE([Average Ticket Price], DIM_PURCHASE_CHANNEL[Railcard] <> "None")
```

Base Measure: [Average Ticket Price]

Comparative Pair: Avg Price (No Railcard)

Avg Price (No Railcard)

01 Passenger Analysis

£#,##0.00

Business Purpose: Computes the average standard ticket price paid by customers without any railcard discount. Allows quantifying the average discount margin provided across railcard programs.

```
CALCULATE([Average Ticket Price], DIM_PURCHASE_CHANNEL[Railcard] = "None")
```

Filter Condition: DIM_PURCHASE_CHANNEL[Railcard] = "None"

Comparative Insight: Direct contrast against discounted cardholders

First Class Mix %

01 Passenger Analysis

0.0%

Business Purpose: Evaluates the proportion of total tickets sold in First Class cabins versus Standard class. Tracks premium passenger uptake and cabin revenue management.

```
DIVIDE(
    CALCULATE([Ticket Count], DIM_TICKET_TYPE[Ticket Class] = "First Class"),
    [Ticket Count]
)
```

Dimension Filtered: DIM_TICKET_TYPE[Ticket Class] = "First Class"

Format: Percentage (1 decimal place)

Advance Booking %

01 Passenger Analysis

0.0%

Business Purpose: Measures the share of tickets booked under Advance fares (requiring booking prior to travel day with seat reservation) versus flexible Anytime or Off-Peak fares.

```
DIVIDE(
    CALCULATE([Ticket Count], DIM_TICKET_TYPE[Ticket Type] = "Advance"),
    [Ticket Count]
)
```

Dimension Filtered: DIM_TICKET_TYPE[Ticket Type] = "Advance"

Operational Impact: Measures passenger planning behavior vs walk-up sales

Online Purchase %

01 Passenger Analysis

0.0%

Business Purpose: Calculates the percentage of tickets purchased digitally via website/app channels compared to in-person station channels. Key metric for digital transformation tracking.

```
DIVIDE(
    CALCULATE([Ticket Count], DIM_PURCHASE_CHANNEL[Purchase Type] = "Online"),
    [Ticket Count]
)
```

Dimension Filtered: DIM_PURCHASE_CHANNEL[Purchase Type] = "Online" Mix Metric: Volume-based channel distribution

Contactless Payment %

01 Passenger Analysis

0.0%

Business Purpose: Tracks the adoption rate of contactless card / mobile tap payments among ticket purchases, reflecting modern frictionless payment preferences.

```
DIVIDE(
    CALCULATE([Ticket Count], DIM_PURCHASE_CHANNEL[Payment Method] = "Contactless"),
    [Ticket Count]
)
```

Dimension Filtered: DIM_PURCHASE_CHANNEL[Payment Method] = "Contactless"

Domain: Payment Operations & Merchant Processing

Peak Travel Band

01 Passenger Analysis

Text / Dynamic String

Business Purpose: Dynamically identifies and outputs the name of the time band (e.g., Morning Peak, Evening Peak, Off-Peak) that experienced the highest volume of departures in context.

```
VAR Bands =
    ADDCOLUMNS(
        VALUES(DIM_TIME[TimeBand]),
        "@Tix", CALCULATE([Ticket Count])
    )
RETURN
    CONCATENATEX(TOPN(1, Bands, [@Tix], DESC), DIM_TIME[TimeBand], ", ")
```

DAX Technique: ADDCOLUMNS + TOPN(1) + CONCATENATEX

Dynamic Context: Adapts instantly to selected date, station, or route filters

02 Sales Performance

Revenue velocities, digital vs station channel splits, daily run-rates, MTD pacing, advance lead times, and same-day booking shares.

Avg Daily Revenue

02 Sales Performance

£#,##0

Business Purpose: Computes the average ticket sales revenue generated per active operating day. Normalizes revenue across months of differing day lengths or partial reporting periods.

```
DIVIDE([Total Revenue], [Journey Days])
```

Dependencies: [Total Revenue], [Journey Days]

Utility: Daily burn-rate and operational revenue run-rate

Best Day Revenue

02 Sales Performance

£#,##0

Business Purpose: Iterates across all calendar dates in context to determine the single highest daily revenue achieved. Useful for identifying peak demand days, holiday surges, and event spikes.

```
MAXX(VALUES(DIM_DATE[Date]), CALCULATE([Total Revenue]))
```

DAX Function: MAXX iterator over distinct dates

Context Transition: CALCULATE triggers context transition per date

Online Revenue

02 Sales Performance

£#,##0

Business Purpose: Total monetary sales revenue collected via digital web and mobile application purchase channels.

```
CALCULATE([Total Revenue], DIM_PURCHASE_CHANNEL[Purchase Type] = "Online")
```

Filter Condition: DIM_PURCHASE_CHANNEL[Purchase Type] = "Online" Referenced By: Online Revenue Mix %

Station Revenue

02 Sales Performance

£#,##0

Business Purpose: Total monetary sales revenue generated through physical station counters and Ticket Vending Machines (TVMs).

```
CALCULATE([Total Revenue], DIM_PURCHASE_CHANNEL[Purchase Type] = "Station")
```

Filter Condition: DIM_PURCHASE_CHANNEL[Purchase Type] = "Station" Channel Strategy: Physical station estate commercial yield

Online Revenue Mix %

02 Sales Performance

0.0%

Business Purpose: Calculates the monetary percentage share of total revenue captured through online channels versus overall network revenue.

```
DIVIDE([Online Revenue], [Total Revenue])
```

Formula: [Online Revenue] / [Total Revenue]

Comparison: Contrasts with Online Purchase % (Volume vs Monetary Mix)

Revenue MTD

02 Sales Performance

£#,##0

Business Purpose: Standard DAX time-intelligence function calculating cumulative Month-to-Date revenue based on the active journey calendar dates.

```
TOTALMTD([Total Revenue], DIM_DATE[Date])
```

DAX Function: TOTALMTD on DIM_DATE[Date]

Pacing Utility: Progress tracking against monthly financial budgets

Avg Purchase Lead Time (Days)

02 Sales Performance

0.0

Business Purpose: Iterates row-by-row through the fact table, decoding integer date keys (YYYYMMDD) into real calendar dates and calculating the mean elapsed days between ticket purchase and actual journey date.

```
AVERAGEX(
    'FACT_TICKET_SALES',
    VAR PKey = 'FACT_TICKET_SALES'[PurchaseDateKey]
    VAR JKey = 'FACT_TICKET_SALES'[JourneyDateKey]
    VAR PDate = DATE(INT(PKey / 10000), MOD(INT(PKey / 100), 100), MOD(PKey, 100))
    VAR JDate = DATE(INT(JKey / 10000), MOD(INT(JKey / 100), 100), MOD(JKey, 100))
    RETURN DATEDIFF(PDate, JDate, DAY)
)
```

Key Parsing: Mathematical extraction of YYYY, MM, DD from integer keys

Strategic Value: Reveals how far in advance passengers plan trips

Same Day Purchase %

02 Sales Performance

0.0%

Business Purpose: Calculates the percentage of tickets bought on the same day as travel (PurchaseDateKey = JourneyDateKey), reflecting spontaneous or commuter booking behavior.

```
VAR SameDay =  
    COUNTROWS(  
        FILTER(  
            'FACT_TICKET_SALES',  
            'FACT_TICKET_SALES'[PurchaseDateKey] = 'FACT_TICKET_SALES'[JourneyDateKey]  
        )  
    )  
RETURN DIVIDE(SameDay, [Ticket Count])
```

Condition: PurchaseDateKey equals JourneyDateKey

Use Case: Last-minute inventory yield management

03 Railway Performance

Operational reliability, punctuality metrics, delay durations (with 24h wrap handling), disruption root causes, and customer refund exposure.

On Time Journeys & On Time %

03 Railway Performance

#,##0 / 0.0%

Business Purpose: Tracks total count and percentage proportion of train journeys that arrived at their destination strictly on time according to the operational schedule.

```
-- On Time Journeys:
CALCULATE([Ticket Count], DIM_JOURNEY_STATUS[Journey Status] = "On Time")

-- On Time %:
DIVIDE([On Time Journeys], [Ticket Count])
```

Benchmark KPI: Network Punctuality Rate

Dimension Filter: DIM_JOURNEY_STATUS[Journey Status] = "On Time"

Delayed Journeys & Delayed %

03 Railway Performance

#,##0 / 0.0%

Business Purpose: Tracks total volume and percentage share of services that completed their route but arrived behind their scheduled arrival timetable.

```
-- Delayed Journeys:
CALCULATE([Ticket Count], DIM_JOURNEY_STATUS[Journey Status] = "Delayed")

-- Delayed %:
DIVIDE([Delayed Journeys], [Ticket Count])
```

Operational Impact: Delay Repay triggering threshold and SLA monitoring

Dimension Filter: DIM_JOURNEY_STATUS[Journey Status] = "Delayed"

Cancelled Journeys & Cancellation %

03 Railway Performance

#,##0 / 0.0%

Business Purpose: Measures the absolute number and percentage rate of booked train journeys that were cancelled prior to reaching the scheduled destination.

```
-- Cancelled Journeys:
CALCULATE([Ticket Count], DIM_JOURNEY_STATUS[Journey Status] = "Cancelled")

-- Cancellation %:
DIVIDE([Cancelled Journeys], [Ticket Count])
```

Regulatory Metric: Rail regulator cancellation penalty index

Dimension Filter: DIM_JOURNEY_STATUS[Journey Status] = "Cancelled"

Avg Delay Minutes

03 Railway Performance

0.0

Business Purpose: Calculates the mean delay duration (in minutes) across all delayed trips. Converts HHMM time keys into elapsed minutes and handles midnight-crossing journeys (where actual arrival is after 00:00).

```
AVERAGEX(
    FILTER('FACT_TICKET_SALES', NOT ISBLANK('FACT_TICKET_SALES'[ActualArrivalTimeKey])),
    VAR SchedKey = 'FACT_TICKET_SALES'[ArrivalTimeKey]
    VAR ActKey = 'FACT_TICKET_SALES'[ActualArrivalTimeKey]
    VAR SchedMins = INT(SchedKey / 100) * 60 + MOD(SchedKey, 100)
    VAR ActMins = INT(ActKey / 100) * 60 + MOD(ActKey, 100)
    VAR RawDiff = ActMins - SchedMins
    VAR AdjDiff = IF(RawDiff < -720, RawDiff + 1440, RawDiff)
    RETURN IF(AdjDiff > 0, AdjDiff)
)
```

Midnight Logic: If difference < -720 mins (crosses midnight), adds 1440 mins

Time Parsing: Decodes HHMM integer keys into total absolute minutes

Total Delay Minutes

03 Railway Performance

#,##0

Business Purpose: Aggregates total passenger-minutes lost to train delays across all journeys in context. Provides the overall scale of operational disruption.

```
SUMX(
    FILTER('FACT_TICKET_SALES', NOT ISBLANK('FACT_TICKET_SALES'[ActualArrivalTimeKey])),
    VAR SchedKey = 'FACT_TICKET_SALES'[ArrivalTimeKey]
    VAR ActKey = 'FACT_TICKET_SALES'[ActualArrivalTimeKey]
    VAR SchedMins = INT(SchedKey / 100) * 60 + MOD(SchedKey, 100)
    VAR ActMins = INT(ActKey / 100) * 60 + MOD(ActKey, 100)
    VAR RawDiff = ActMins - SchedMins
    VAR AdjDiff = IF(RawDiff < -720, RawDiff + 1440, RawDiff)
    RETURN IF(AdjDiff > 0, AdjDiff, 0)
)
```

Aggregate Function: SUMX iterator over non-blank actual arrivals

Output: Cumulative delay minutes formatted as whole integer

Refund Requests, Refund Rate % & Revenue at Risk

03 Railway Performance

#,##0 / 0.0% / £#,##0

Business Purpose: Analyzes passenger compensation claims resulting from disruptions. Measures total claim count, claim rate relative to total tickets, and gross monetary value at risk of refund pay-out.

```
-- Refund Requests:
CALCULATE([Ticket Count], 'FACT_TICKET_SALES'[Refund Request] = "Yes")

-- Refund Request Rate %:
DIVIDE([Refund Requests], [Ticket Count])

-- Revenue at Risk (Refunds):
CALCULATE([Total Revenue], 'FACT_TICKET_SALES'[Refund Request] = "Yes")
```

Fact Column Filtered: FACT_TICKET_SALES[Refund Request] = "Yes"

Financial Impact: Quantifies financial loss and compensation exposure

Top Delay Reason

03 Railway Performance

Text / Dynamic String

Business Purpose: Identifies the primary operational cause (e.g. Signal Failure, Weather, Technical, Staffing) responsible for the greatest frequency of passenger delays in the current filter context.

```
VAR Reasons =
    FILTER(
        ADDCOLUMNS(
            VALUES(DIM_JOURNEY_STATUS[Reason for Delay]),
            "@Cnt", CALCULATE([Ticket Count])
        ),
        NOT ISBLANK(DIM_JOURNEY_STATUS[Reason for Delay])
    )
RETURN
    CONCATENATEX(TOPN(1, Reasons, [@Cnt], DESC), DIM_JOURNEY_STATUS[Reason for Delay], ", ")
```

Blank Handling: Excludes BLANK/On-time entries to isolate true delay reasons

Ranking: TOPN(1, ..., DESC) selects highest ticket disruption factor

04 Route Analysis

Station-to-station corridor metrics, network yields, destination role-playing analytics, and top route identification.

Distinct Routes, Avg Tickets & Avg Revenue per Route

04 Route Analysis

#,##0 / #,##0.0 / £#,##0

Business Purpose: Calculates the number of unique origin-destination station pairs and determines the average passenger volume and revenue generated per route corridor across the rail network.

```
-- Distinct Routes:
COUNTROWS(
    SUMMARIZE(
        'FACT_TICKET_SALES',
        'FACT_TICKET_SALES'[DepartureStationKey],
        'FACT_TICKET_SALES'[ArrivalStationKey]
    )
)

-- Avg Tickets per Route:
DIVIDE([Ticket Count], [Distinct Routes])

-- Avg Revenue per Route:
DIVIDE([Total Revenue], [Distinct Routes])
```

Summarization: Origin-Destination key pairs grouped via SUMMARIZE

Network KPI: Network corridor density and route performance benchmarks

Revenue & Tickets by Arrival Station

04 Route Analysis

£#,##0 / #,##0

Business Purpose: Evaluates revenue and ticket counts grouped by the *Destination / Arrival Station* rather than the origin station by activating the inactive relationship between the fact table and DIM_STATION.

```
-- Revenue by Arrival Station:
CALCULATE(
    [Total Revenue],
    USERELATIONSHIP('FACT_TICKET_SALES'[ArrivalStationKey], DIM_STATION[StationKey])
)

-- Tickets by Arrival Station:
CALCULATE(
    [Ticket Count],
    USERELATIONSHIP('FACT_TICKET_SALES'[ArrivalStationKey], DIM_STATION[StationKey])
)
```

Role-Playing Activation: USERELATIONSHIP on ArrivalStationKey -> StationKey

Analytical Value: Inbound passenger flow analysis for destination stations

Top Departure Station

04 Route Analysis

Text / Dynamic String

Business Purpose: Dynamically calculates and returns the name of the origin station originating the greatest ticket volume in context.

```
VAR Stations =
    ADDCOLUMNS(
        VALUES(DIM_STATION[StationName]),
        "@Tix", CALCULATE([Ticket Count])
    )
RETURN
    CONCATENATEX(TOPN(1, Stations, [@Tix], DESC), DIM_STATION[StationName], ", ")
```

Dimension Column: DIM_STATION[StationName]

Utility: Executive scorecard callouts and station hub ranking

Busiest Route

04 Route Analysis

Text / Dynamic String

Business Purpose: Finds the single origin-destination station pair with the highest ticket volume and formats the corridor into a human-readable text string (e.g. "London Euston to Manchester Piccadilly").

```
VAR Routes =
    ADDCOLUMNS(
        SUMMARIZE(
            'FACT_TICKET_SALES',
            'FACT_TICKET_SALES' [DepartureStationKey],
            'FACT_TICKET_SALES' [ArrivalStationKey]
        ),
        "@Tix", CALCULATE([Ticket Count])
    )
VAR TopRoute = TOPN(1, Routes, [@Tix], DESC)
RETURN
    CONCATENATEX(
        TopRoute,
        LOOKUPVALUE(DIM_STATION[StationName], DIM_STATION[StationKey],
            'FACT_TICKET_SALES' [DepartureStationKey])
        & " to "
        & LOOKUPVALUE(DIM_STATION[StationName], DIM_STATION[StationKey],
            'FACT_TICKET_SALES' [ArrivalStationKey]),
        ", "
    )
```

Corridor Lookup: Uses LOOKUPVALUE twice to resolve origin & destination station names Output String: "[Origin Station] to [Destination Station]"

Route Yield per Ticket

04 Route Analysis

£#,##0.00

Business Purpose: Evaluates revenue per ticket in route-level visual contexts. Used to spot high-yield premium corridors versus high-volume commuter corridors.

DIVIDE([Total Revenue], [Ticket Count])

Financial Application: Route profitability & corridor pricing optimization

Format: Currency (£#,##0.00)

APPENDIX DAX Modeling Patterns & Reference Matrix

Pattern / Technique	Measures Using Technique	Architectural Rationale & Best Practice
Safe Division (DIVIDE)	[Average Ticket Price], [Revenue MoM %], [On Time %], [Delayed %], [Cancellation %], [Refund Request Rate %], [Advance Booking %]	Prevents #DIV/0! errors when filtered to zero volume and returns BLANK() cleanly.
Role-Playing (USERELATIONSHIP)	[Total Revenue (by Purchase Date)], [Revenue by Arrival Station], [Tickets by Arrival Station]	Avoids duplicating physical dimension tables in Power BI; dynamically activates secondary relationships in DAX calculation context.
Iterators (SUMX / AVERAGEX)	[Avg Purchase Lead Time (Days)], [Avg Delay Minutes], [Total Delay Minutes]	Calculates custom row-level logic (e.g. date parsing, 24h midnight-crossing adjust) without bloating fact table storage with calculated columns.
Dynamic String Ranking (TOPN + CONCATENATEX)	[Peak Travel Band], [Top Departure Station], [Top Delay Reason], [Busiest Route]	Generates dynamic narrative card KPI callouts that automatically respond to all slicers and cross-filters.
Time Intelligence (PREVIOUSMONTH, TOTALMTD)	[Revenue Previous Month], [Revenue MoM %], [Ticket Count MoM %], [Revenue MTD]	Relies on a contiguous Date table (DIM_DATE) to calculate standard month-over-month and cumulative pacing KPIs.

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_Measures Table